C-DAC Mumbai
Lab Assignment

Problem 1: Grade Evaluation System

```
import java.util.*;

class grade{
	public static void main(String args[]){
		Scanner sc = new Scanner(System.in);
		System.out.print("Enter Marks in Maths : ");
		int m = sc.nextInt();
		System.out.print("Enter Marks in Science : ");
		int s = sc.nextInt();
		System.out.print("Enter Marks in History : ");
		int h = sc.nextInt();

		int avg = (m + s + h)/3;
		System.out.println("Average Marks : " + avg);

		if( avg >= 90){
			System.out.print("Grade A");
		}
		else if( (avg >= 70) && (avg <= 89)){
			System.out.print("Grade B");
		}
		else if((avg >= 50) && (avg <= 69)){
			System.out.print("Grade C");
		}
		else if((avg >= 30) && (avg <= 49)){
			System.out.print("Grade D");
		}
		else{
			System.out.print("Fail");
		}

}
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>javac grade.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>java grade
Enter Marks in Maths : 80
Enter Marks in Science : 85
Enter Marks in History : 90
Average Marks : 85
Grade B
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>java grade
Enter Marks in Maths : 95
Enter Marks in Science : 90
Enter Marks in History : 98
Average Marks : 94
Grade A
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>
```

Problem 2: Leap Year

```java
import java.util.*;

class LeapYear{
	public static void main(String args[]){
		Scanner sc = new Scanner(System.in);
		System.out.print("Enter year : ");
		int y = sc.nextInt();

		if(((y % 4 == 0) && (y % 100 != 0)) || (y % 400 == 0)){
			System.out.print(y + " is a Leap Year ");
		}
		else{
			System.out.print(y + " is not a Leap Year ");
		}
}
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>javac leapyear.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>java LeapYear
Enter year : 2024
2024 is a Leap Year
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>java LeapYear
Enter year : 1900
1900 is not a Leap Year
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>
```

Problem 3: Days of the Week

```java
import java.util.*;

class Dow{
	public static void main(String args[]){
		Scanner sc = new Scanner(System.in);
		System.out.print("Enter day number : ");
		int d = sc.nextInt();
```

```
switch(d){
        case 1: System.out.print("Monday"); break;
        case 2: System.out.print("Tuesday"); break;
        case 3: System.out.print("Wednesday"); break;
        case 4: System.out.print("Thursday"); break;
        case 5: System.out.print("Friday"); break;
        case 6: System.out.print("Saturday"); break;
        case 7: System.out.print("Sunday"); break;
        default: System.out.print("Invalid day number"); break;
}
}
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>javac daysofweek.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>java Dow
Enter day number : 3
Wednesday
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>java Dow
Enter day number : 0
Invalid day number
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>java Dow
Enter day number : 5
Friday
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>
```

Problem 4: Identify the Values of Uninitialized Variables

```
class demo{
  public static void main(String args[]){
    byte a = 25;
    short b = 100;
    int c = 1000;
    long d = 10000L;
    float e = 10.5f;
    double f = 20.55;
    char g = 'A';
    boolean h = true;

    System.out.println(a);
    System.out.println(b);
    System.out.println(c);
    System.out.println(d);
    System.out.println(e);
    System.out.println(f);
    System.out.println(g);
    System.out.println(h);
}
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 2\Assignment>java demo
25
100
1000
10000
10.5
20.55
A
true
```